Md. Abdullah Al-Fahim

Civil Engineering Researcher | Computational Geomechanics

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Research Summary

Civil engineer working between experimental soil mechanics and computational modelling. I build **discrete element models** of granular soils from scratch and calibrate them against matched laboratory tests, currently on how the inherent (bedding-plane) anisotropy of sand carries over to sand–structure interfaces. I am seeking a graduate (**MSc/PhD**) research position in computational geomechanics and related areas.

Interests: discrete element method; inherent anisotropy and fabric; soil–structure interfaces; contact-model calibration and validation; specimen-size effects in direct shear; sustainable construction.

Education

Rajshahi University of Engineering & Technology (RUET) **2019–24**

Bachelor of Science in Civil Engineering *Rajshahi, Bangladesh*

- **CGPA**: 3.06/4.00

Research Experience

Undergraduate Thesis **2023–24**

Dept. of Civil Engineering, RUET *Supervisor: Prof. Md. Mahmud Sazzad*

"Influence of Inherent Anisotropy of Soil on the Behavior of Sand-Concrete Interface"

- Developed a custom two-dimensional, **GPU-accelerated Discrete Element Method (DEM) engine** from scratch in Python/PyTorch (float64) to simulate direct shear of clumped-grain assemblies, using elastic–perfectly plastic tangential and rolling-resistance contact laws and a sawtooth concrete interface.
- Designed a matched laboratory programme of **48 direct shear tests** (ASTM D3080) on a river sand and a crushed sand, sheared against themselves and against concrete, at four bedding angles and three normal stresses, with three random-seed replicates per simulated condition.
- Applied a strict **calibrate-once protocol**: contact parameters tuned against experiment at a single bedding angle and then frozen, so that strength at all other bedding angles is a blind prediction rather than a re-fit.
- Analysed how inherent (bedding-plane) anisotropy is transmitted from sand–sand shearing to the sand–concrete interface, combining stress–strain response with contact-normal, force-weighted and grain-orientation fabric tensors.
- Studied the effect of specimen size on whether bulk anisotropy is captured in direct shear, and used the outcome to select the specimen geometry for the interface simulations.
- Journal manuscript based on the thesis is **in preparation** (see Publications).
- Next: extending the framework to sand–pile interfaces under lateral loading.

Barriers to Green Buildings in Bangladesh | *Survey, Analytic Hierarchy Process* **RUET, 2024**

- Survey and Analytic Hierarchy Process analysis of the hindrances to green building construction ($n = 87$ responses; consistency ratio 0.011); **government involvement ranked first (25.0%)**. First-author conference paper (ICCESD 2024).

Publications

Journal Articles

- **Al-Fahim, M.A.**, Sazzad, M.M., Shafi, M.S.M. *Transmission and Attenuation of Bedding-Plane Anisotropy from Sand to Sand-Concrete Interfaces: A Discrete Element Study.* **In preparation.**

Conference Papers

- **Al-Fahim, M.A.**, Shahriar, S. *Hindrances in Constructing Green Buildings in Bangladesh: A Comprehensive Study.* 7th International Conference on Civil Engineering for Sustainable Development (**ICCESD 2024**), KUET, Khulna, Bangladesh.

Technical Skills

Computational: Python (PyTorch, custom DEM/FEM code), R, ABAQUS

Laboratory: direct shear testing (ASTM D3080), particle-size analysis, concrete mix design (ACI 211)

Methods: calibration and validation of numerical models, survey design, Analytic Hierarchy Process

Design & Productivity: AutoCAD (2D), SketchUp, Lumion; MS Word, PowerPoint, Excel

Professional Experience

RANCON Engineering & Consultants Ltd **Nov. 2025 – Present**

Executive (QC-PMO) *Tejgaon, Dhaka*

- Inspect structural, architectural and finishing works and incoming materials against approved drawings, specifications, BOQ and BNBC; resolve non-conformities with contractors and keep QA/QC records (checklists, NCRs, corrective actions).
- Independently developed a live internal **portfolio-tracking dashboard** (Python/Flask) integrating project schedules with progress, dependency and financial reporting across concurrent developments.
- Developed and piloted **RANCON QMS**, a web-based digital inspection system: digital QC checklists, a raise–respond–verify–close NCR workflow, a role-based approval chain (Project Engineer, Project In-charge, HSE, QC/PMO), photo and PDF attachments, and in-app notifications.

Mir Cement Ltd. **Jan. 2025 – Nov. 2025**

Engineer (Civil) *Rupganj, Narayanganj*

- Developed and monitored project schedules and prepared cost estimates, budgets and resource plans.
- Designed and optimised concrete mixes per ACI 211.R and verified raw materials and finished concrete against quality standards.

Presentations & Activities

- Poster Presentation — Retrofitting Old Toilets for Sustainable Bangladesh (**WaterAid Toilet Conference, 2025**)
- Team Event — Game of Trusses (**Ceenovation, IUT, 2024**)
- Poster Presentation — Sustainable Microgrids: Enhancing Campus Energy Resilience (**Technofair, RUET, 2023**)

References

- **Dr. Md. Mahmud Sazzad** (thesis supervisor) - mmsruet@gmail.com - Professor, Dept. of Civil Engineering, RUET - +8801744355019
- **Md. Rashidul Islam** - rashidul.islam@ranconrecl.com - Head of Department (PMO), Rancon Engineering and Consultants Ltd. - +8801329702931